Understanding the AR-VR Based Architectural Design Workflow among Selected Danish Architecture Practices

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Virtual reality (VR) and augmented reality (AR) have been proposed to be additional architectural design mediums for at least 25 years (Dagit, 1993). Despite rapid technical and technological development, it has not been adopted into architectural design practices as compared to academia and research. Surveys from the American Institute of Architects (AIA) and Royal Institutes of British Architects (RIBA) demonstrate the state of architectural practices; 72% of architects and 65% of architects respectively are not using any kind of virtual, augmented, or mixed reality in their practices(RIBA and Microsoft, 2018; Hampson, 2020). In this paper, the authors investigate the state of practices, issues, challenges, and opportunities of the utilization of virtual, augmented, and mixed realities in six architectural practices in the Danish context. Three of the practices are large architectural practices, one medium-sized practice specializing in institutional, healthcare and cultural architecture, and one firm designing private family houses, kindergartens, daycares and places for people with disability and, one experimental design studio. All these practices have used VR/AR in their projects to various degrees. In recent years Danish architectural practices have been involved in various VR/AR-based exhibitions, demonstrations, and tool developments to promote the usage of the same in design practice. Through a set of qualitative interviews with personnel from key architectural practices, the authors would like to demonstrate the present state of practices. The investigation explores the usage of VR and AR in Danish architecture practices by identifying challenges and opportunities regarding skill levels, architectural typology, use cases, toolchains, and workflow and shows similarities and differences between traditional and VR-based design processes. The main findings show how VR/AR-based visualization helps architects to perceive spatiality and also ushers creativity through immersion and overlays.

Keywords: *Virtual Reality, Augmented Reality, Architectural Design Practice, Denmark.*

INTRODUCTION

In his 1935 novel, "Pygmalion's Spectacles", Stanley G. Weinbaum (Weinbaum, 2013), had created a scenario where an artificial reality was not only possible through visual and auditory senses but all five senses, which was one of the very first fictional existence of virtual reality(VR). Since then through technological advances, many of these fictional technologies have come to life and have been well integrated into certain professional practices. Since the start of the decade 1990, it has been speculated, delineated and, implemented about how to utilize these technologies in architectural design processes (Dagit, 1993). But due to the speed of the

development of the technology at the hardware level and its complex implementation into the well-established workflow of architectural design, it has been poorly adapted to the practice of architecture. After the pandemic situation and thirty-three years after coining the term "Virtual Reality"(Steuer, 1992), the authors would like to investigate the use and integration of VR in architectural practices prevalent within Denmark.

Former studies in this field have so far concentrated on architectural design as a discipline. Studies by de Freitas and Ruschel(Regina de Freitas and Coeli Ruschel, 2013) show the research and development of the application of augmented reality (AR) and virtual reality (VR) by reviewing 200 papers from various conferences over 11 years. But this focuses majorly on academic research and development rather than practice-based scenarios. A study by Delgado et al. (Davila Delgado et al., 2020) shows data from a research network comprising academic and professional partners regarding the use of VR and AR. This shows results correlating both to academia and practice while concluding that VR and AR are not quite ready to be assimilated into architectural practice, yet.

Another literature review by Adeline Stals and Luisa Caldas (Stals and Caldas, 2020) has demonstrated only a mere 2% of all the research paper reviews (4 out of 201) in their purview, contained relevant feedback from the architects.

A potential solution to address this research gap was relatively simple; interviewing practising architects who work integratively with VR and/or AR. Through these interviews, it was sought to address the present state of affairs, challenges, opportunities and scope of development to integrate VR and/or AR into the practice in a more holistic manner. The conducted interview could have been qualitative and/or quantitative, but for this particular study, qualitative interviews have been conducted to take into the context and to understand the issues (Kvale, 2007)

This paper will present the research findings of the interviews about the implementation of VR and AR into architectural design practice. It will also present (1) the reason for the selection of the architectural design practices based on their varied profile and global and local presence, and (2) their methodology and involvement with the VR/AR technology with existing and custom workflow and toolchain.

RESEARCH METHODS

This research employs qualitative interviews and ethnographic research as the core of the research design. To understand the usage of VR/AR in architectural design practice it is important to know the working habits and trends of the architectural design studios. Ethnographic studies were conducted to understand this part. Following this study, a pre-designed questionnaire was used to know more about the background, present state and future ambitions of VR/AR-based architectural design practices.

Before moving towards the interview, its design and result, it is imperative to understand the participants and the reason for choosing them. The interview consisted of six different architectural practices of varied scale and demeanour. In common interview studies, the number of participants usually ranges from 5 to 25. Particularly qualitative studies, it is often designed on the paradigm of design of quantitative studies where more interview correlates to more scientific results. In this case, the need was to understand the qualitative practice-based understanding, which was achieved by the range of practice. (Kvale, 2007)

Participant Selection

First, the six architectural design practices can be categorised into two broad categories. (1) large architectural practices; Henning Larsen Architects, Schmidt Hammer Lassen Architects, AART and (2) small to medium range architectural practices; Friis and Moltke A/S, Arkitektfirmaet Hovadt, and POINT.

In the first category, Henning Larsen Architects presently operates on three continents in seven different offices. Since 2019, they have also joined

Ramboll to extend their portfolio. Schmidt Hammer Lassen operates only in Denmark in 2 offices and one in Shanghai. But they are part of Perkins and Will since 2018. AART identifies itself as a Scandinavian design practice, having four offices in Denmark, Norway and Sweden.

In the second category, Friis and Moltke A/S is a medium-sized architectural practice, operating mostly in Denmark from the North Jutland region, working with healthcare, cultural and institutional projects. Arkitektfirmaet Hovadt is a small practice, also from North Jutland, region working with small to large residential projects, institutions, and businesses. POINT is an experimental architectural office from Copenhagen that has designed experimental installations, small houses with unusual requirements and landscape architecture based projects.

This particular choice was made to represent architectural design practices with rich Scandinavian design heritage to new experimental works, contributing to international, Scandinavian and Danish architecture. Despite their wide range of operations and architectural expressions, all these firms have been using virtual reality in the architectural design process for a minimum of three years to a maximum of six years.

Interview Structure

The qualitative questionnaire was created on the premise of their use of VR in the architectural design process. Hence, no questions have been formulated to investigate the reason for not using VR in their design pipeline.

The interview was designed to gather as much data possible from their workflow and usage of VR in the practice and later find the common pattern from the information obtained. The queries in the questionnaire were designed to find the following: (1) timeline of the usage of VR, (2) possible use of AR, (3) types of projects in which VR was implemented, (4) hardware and software, (5) design stages of deployment, (6) knowledge creation in VR, (7) details about involved personnel and skill levels, (8) possible use of game engines, (9) challenges of utilization: both methodology wise and logistics wise, and (10) preferred future development track.

RESEARCH IMPLEMENTATION

A survey (Firm Survey Report 2020) from the American Institute of Architects(AIA) shows that only 28% of the architectural firm registered under AIA uses VR at some level in their design pipeline(Hampson, 2020). A similar survey from the Royal Institute of British Architects shows a similar picture where only 35% of offices use VR or AR or any sort of Mixed Reality(MR) tool in their design pipeline(RIBA and Microsoft, 2018). It can be argued that with the rapid development of tools and democratization of technologies in between these years, these numbers have already been outdated. During this timeline, a global pandemic also happened, meaning it would make sense to move the task of architectural design into a virtual world. Bjarke Ingels Group and UNStudio, during this time, have tried to design a VR/AR-based design tool called Hyperform (Squint/Opera, BIG and UNStudio create virtual-reality design tool, 2022). In reality, it ended up being a closed-source presentation tool with a markup facility (SpaceForm-Immersive Presentations for Global Audiences, 2022) .Also, the Danish Association of Architects (Arkitektforeningen), which is operational since 1879 did not have this data. At this stage, without the availability of further data, it was necessary to interview architects on this scale to generate qualitative data about the state of affairs and to understand the need for the same in the profession.

Interview Details

Six architects from these six practices were interviewed. Their designation ranges from the founder, director, partner and senior architect. Given their experience level and involvement with the architectural design practice, they are all involved in the design process actively. Moreover, they are all involved in the VR based design and design review process.

Questionnaire

During the interview, the participants were asked the following details which eventually contributed to the knowledge base. (1) At which type of project and at which design phases VR has been used, (2) since how long they have been using the VR setup, (3) who are the major stakeholders in the design and demonstration process and how those knowledge cases were developed, (4) what is their main CAD, 3D, BIM and Visualization packages and how VR tools and hardware are linked to it. (5)if they use any game engine and if so how that is connected to their usual workflow, (6) if they still use a physical model in tandem with their digital and VR models and how does that work, (7) how a spatial understanding is changed through VR, (8) what is the level of their infrastructure regarding the VR and (9) what do they expect from VR based development in future regarding architectural design.

Research Focus

From the data from AIA and RIBA (RIBA and Microsoft, 2018; Hampson, 2020) it was prevalent that VR and AR to some extent are mostly used for presentation and marketing purposes. These usages often outweigh the purpose of design. The data from the interview was initially checked for consistency with this conclusion. It was studied further to extract relevant information specific to our queries.

RESULT

The results from the interviews yielded insights into the workflow. They are explained in detail in the following sections.

Reason for using VR

For different practices, this yields different answers. In the case of POINT, and Arkitektfirmaet Hovadt(AH), it was a demand from the client as they needed to experience the space for their preference and accessibility. In the case of Henning Larsen(HL), Schmidt Hammer Lassen(SHL) and AART it was for both marketing purposes, a new mode of communication with the client and a new mode of design experience for the architects. In the case, of Friis and Moltke(FM), it was a motivation to share the visualization in a new way with its stakeholders within the project budget along with a new ground to discuss. But eventually, VR has developed into a design experience tool for all practices and it has been involved since the concept design and massing stage.

Another common denominator for these practices is, that none of the practices "felt" the need for augmented reality. Also, the steep price range for AR headsets play a role in this matter.

Skillset for VR

In Denmark, architectural education can be obtained only from 3 schools: Royal Danish Academy, Aarhus School of Architecture and Aalborg University. None of the schools has any program or course at either bachelor's or master's level that teaches the relevant skill to use VR in architecture. The people using these tools in these offices are self-trained over some time and new architects in the offices are trained similarly through internal workshops, peer-to-peer learning or the application. Recently, the rise of academic licenses for certain tools is enabling self-motivated students to learn these tools according to the industry need before coming into the design industry.

Timeline and correlation with software and hardware

In this particular survey, HL is the oldest participant in the VR based architectural design practice. They started this practice almost six years ago with Samsung VR(Samsung Gear VR with Controller - The Official Samsung Galaxy Site, 2022). This was used in tandem with 360 degree renders from render engines and some specific steps to create this experience. Now HL has shifted to Enscape (EnscapeTM - Real-Time Rendering and Virtual Reality, 2022), a real-time visualization engine that offers VR functionality.

Table 1
Timeline, Design Stages and Mode of use

	Years in Practice	Design Stage	Mode of Use
Henning Larsen	5-6	Schematic Design Design Development Visualization	Immersive Model 360° View
Schmdit Hammer Lassen	2-3	Schematic Design Design Development	Immersive Model
AART	5-6	Schematic Design Design Development	Immersive Model
POINT	4	Schematic Design Design Development Visualization	Immersive Model
Arkitektfirmaet Hovadt	3	Schematic Design Design Development Visualization	Immersive Model
Friis & Moltke	4-5	Schematic Design Design Development Visualization	Immersive Model

Table 2
Tools and Hardware

	Hardware	Drawing + 3D Software	VR Authoring Tool
Henning Larsen	Meta Quest Oculus Rift	AutoCAD-Rhino3d Revit	Enscape
Schmdit Hammer Lassen	Oculus Rift HTC Vive	AutoCAD-Rhino3d Revit	Enscape
AART	Oculus Rift	AutoCAD-Rhino3d + SketchUp Revit	Enscape
POINT	HTC Vive	AutoCAD + SketchUp	Enscape Unreal Engine
Arkitektfirmaet Hovadt	HTC Vive	ArchiCAD	Twinmotion
Friis & Moltke	Oculus Rift	AutoCAD-Rhino3d + SketchUp Revit	Enscape

POINT started with Unity four years back to create an experience for a client but due to ease of use, they have also presently switched to Enscape. This switch has been made due to the real-time render capability of Enscape. In the case of AH, they are using Twinmotion(A Cutting Edge Real-Time Architectural Visualization Tool - Twinmotion, 2022) for the same real-time render capability but also its integration with Graphisoft ArchiCAD(Archicad – Graphisoft, 2022).

All the other practices use either SketchUp(3D Design Software | 3D Modeling on the Web | SketchUp, 2022) or Rhino3D(Rhino - Rhinoceros 3D, 2022) as their 3D editor and Revit(Revit Software | Autodesk, 2022) as BIM tool. As enscape offers a plug-in to all the three mentioned software and a VR functionality it is the natural choice for the rest.

But, by using Enscape, all these practices are limited to now Oculus Rift(now obsolete) and HTC Vive. Though Oculus Rift has been superseded by Oculus Quest 2(Oculus Quest 2: Our Most Advanced New All-in-One VR Headset | Oculus, 2022), users are forced to use the tethered Oculus Rift with "outside-in" tracking(VR positional tracking - Wikipedia, 2022). In the case of HTC Vive, it is well supported by the company but its comparative steep price and tethered functionality limit the architects from investing in it.

VR in stages of Architectural design

In all the interviews one issue became prominent, though the practice had started as a visualization tool, soon everyone adopted it as a design sketching tool in the conceptual development stage as it offers a better sense of space compared to physical models and drawing. In Albena Yaneva's ethnographic work with OMA(Yaneva, 2009) it is mentioned that physical models provide a much more sense of space as compared to a digital model on a screen. But given the fact, that the work was done in 2002, twenty years before now, the same hypothesis might not hold for these offices. Also, for all of these offices, a physical model is time-consuming and expensive to build and that cost is not feasible in the early stages of the project.

Also, all the subjects in the interview have drawn a consensus on the fact that in a VR environment, communication is easy among an inexperienced intern, a moderately experienced junior architect, an experienced senior architect and, often stakeholders who do not always understand the classical representation of architecture through technical drawings like a plan, section, elevation or 2D visualization. For POINT specifically, they use VR as a method of implementing an "artificial memory" of a

space in a person's memory where the space does not physically exist yet.

Due to these advantages, which far outweigh the technical challenges of setting up a VR with the limited setup opportunities, architects prefer to adopt VR in the early stages of design and continue in the design development.

Present Challenges

The present challenges to the architects can be classified into the following categories.

- One has to be dependent on a real-time visualization engine to be able to use VR easily and fast.
- Not all offices can afford to dispense certain office hours toward the development of new custom tools which will allow them to use VR bypassing these visualization engines.
- Due to a mismatch between old hardware, new software and incompatible tethered PC, clients have often become nauseated which breaks the flow of discussion and creates uneasiness in the studio.
- Some of these old headsets need "base stations"(Base Station | VIVE European Union, 2022) which occupy certain square meters of space that none of the offices can afford to have ready all the time so they prefer a wireless solution like Quest 2. But often that does not become reality due to limitations imposed by the visualization engines.

Future Expectations

Based on these interviews conducted, for the future development of technologies and capabilities, all the architecture practices draw another consensus.

- Like CAD and BIM, VR based architectural design education needs to be incorporated into architecture schools.
- There is a need for a software bridge or a framework that allows architects to send 3D model and simulation data in a wireless headset bypassing the limitations of the visualization engine. For augmented reality-based headsets, Fologram(Fologram, 2021) has already implemented the same. But similar VR based application needs to be developed.
- VR in itself is a very "introverted" media. Thus, the need to collaboratively view and edit the same model is of paramount importance.
- Presently the decisions made while viewing the space in VR are marked manually either in a model or a drawing. The practices would like to mark these annotations in VR also to keep the creative flow and gestural communication intact.
- Recent developments in software like Arkio allow multiple architects to connect in a shared VR model using modern wireless VR headsets and allow for drawing and annotation. These tools often operate as a plug-in to Rhino, SketchUp, Revit and also Unity. A newly designed workflow through these tools needs to be introduced in the studios in their existing workflow.

From the acquired results, it is observed that the practices draw consensus on certain matters. These are mostly technological and logistical barriers and opportunities regarding VR. But in their individual workflow, they differ in the following aspects: (1) use of certain tools, (2) types of projects in which VR is used, and (3) utilization of the technology. These differences will exist as long as there are different approaches to a design. Nevertheless, the technology should be flexible enough to cater to all these differences.

CONCLUSION

This study scratches the surface of an issue faced globally. This study is based on some local architecture practices and some Denmark based practices which operate globally. To get more relevant data and to address the technological aspect as well as the sociological aspect of these

technologies, the study needs to be extended to more countries. There is a further scope and need to study the architectural design practices, that do not use VR / AR / MR in their workflow. As these technologies are developing at a rapid and fast pace and being adopted into various other practices, it will be imperative to understand the counter logic of those practices regarding their non-inclusion.

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